

Physical Frictions and Stylized Facts

(Stylized Facts Asset Returns, Group 18)

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1 Introduction, Motivation and Method

A return is defined as the change in price over a specific time period. When investing the general goal is to have that percentage value be positive (making profit). Asset returns exhibit numerous characteristic properties, which warrants closer analysis of the data’s statistical behavior.

This report seeks to understand what actually drives extreme market risks. Constructing a “Volatility Spectrum” will allow us to examine how stylized facts (heavy tails, skewness, aggregational Gaussianity, volatility clustering, and the leverage effect) differ between high and low volatility sectors. Beyond that we will explore whether the differences stem from physical constraints or financial mechanics.

Risk Profile	Physical Frictions	Asset-Light (Digital)
High Risk	Energy (Enrgy)	Technology (HiTec)
Low Risk	Consumer Goods (NoDur)	Telecom (Telcm)

Table 1: Matrix of risk, with Healthcare (Hlth) acting as a baseline.

In order to carry out our analysis, we extracted five specific sector portfolios, from the “10 Industry Portfolios [daily]” dataset over a 10-year horizon (Kenneth R. French Data Library).

2 Empirical Analysis: The Stylized Facts

2.1 Heavy Tails

A fundamental stylized fact of financial asset returns is the presence of heavy tails indicating that extreme positive and negative returns occur far more frequently in practice than what is predicted by a standard Gaussian (normal) distribution. Under a standard

normal distribution, the probability of encountering an event beyond three standard deviations ($\pm 3\sigma$) is less than 0.3%, while a six-standard-deviation event ($\pm 6\sigma$) is practically impossible (occurring roughly once every 4 million years), called a "Black Swan" event. We examined kurtosis by the formula:

$$\kappa = \frac{E[(R_t - \mu)^4]}{\sigma^4}, \quad \kappa_{excess} = \kappa - 3$$

Where the values for the excess kurtosis are seen in table 3. Any distribution with excess kurtosis above 0 is seen as *leptokurtic*, implying high peakiness around the mean and much heavier tails.

	Enrgy	HiTec	NoDur	Telcm	Hlth
Excess Kurtosis	14.06717	8.788897	16.07018	8.904935	7.848258

Table 2: Excess kurtosis of daily returns

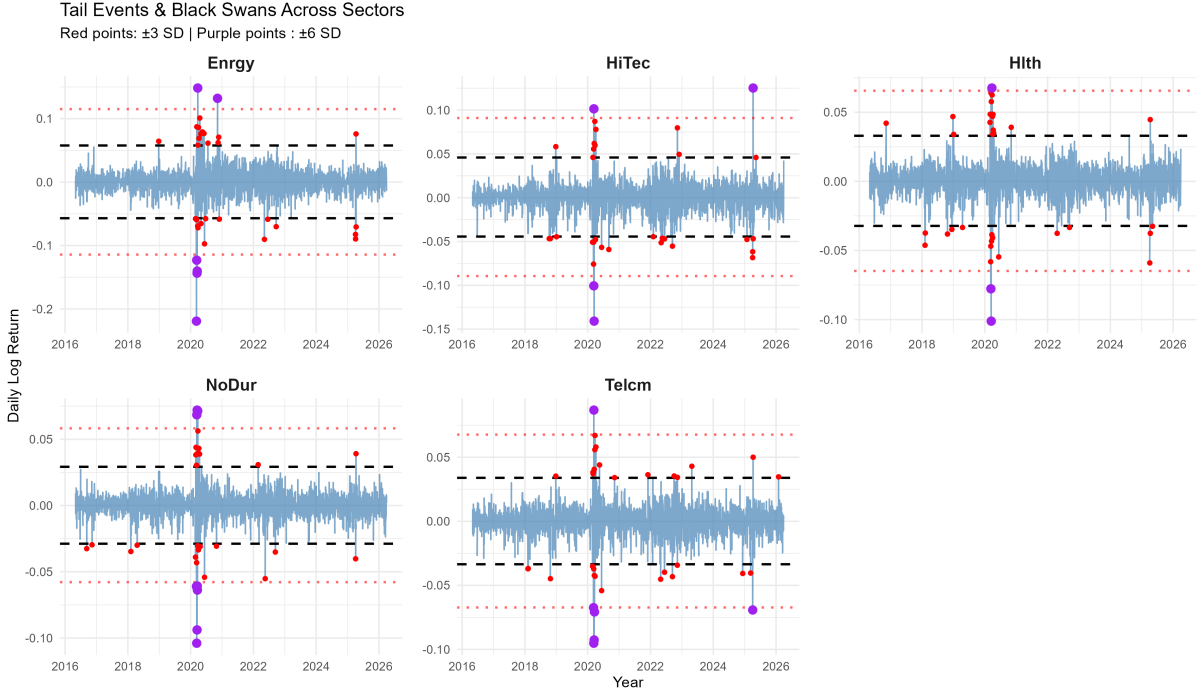


Figure 1: Tail event frequency across sectors

At high intervals (daily), short-term tail risk is highly present, inversely correlated to the length of the time frequency, but varies structurally across industries. The daily (and weekly) return distributions for all five sectors are heavily leptokurtic, indicating a universal exposure to extreme price shocks that far exceed the predictions of a normal distribution. Non-Durables (16.07) and Energy (14.07) exhibit the most severe daily excess kurtosis, highlighting vulnerability to sudden disruptions like supply chain bottlenecks or geopolitical shocks. High Tech (8.79) and Telecom (8.90) similarly display high daily fat tails, likely driven by rapid information arrival, earnings calls (and surprises), and speculative trading in growth-oriented equities, especially with the rapid growing availability for people to engage individual trading. Speculative technology and telecom stocks, such as Nvidia, Sandisk, or ASTS see inflated trading volumes as traders

speculate on the growth of the rapidly scaling industries. In contrast, the Healthcare sector demonstrates the lowest daily (7.85) and weekly (2.64) excess kurtosis, protected from short-term market re-pricings; in line with our hypothesis of a constant demand for healthcare dampening the industry volatility.

Based on figure 1, we can see that the frequency of $\pm 3\sigma$ events are higher than in a normal distribution. Assuming an occurrence rate of 0.3% of a $\pm 3\sigma$ event, we calculate that over a 10 year period, there are 2520 trading days, of which 7 would be therefore a $\pm 3\sigma$ event, much higher than the dozens of red events over/under the black dashed lines we see for every sector. The same applies for $\pm 6\sigma$ events, assuming a 0.0000000002% probability, over 2520 trading days there would be 0 black swan events. In reality, there are for every sector 3-5 purple dots over/under the $\pm 6\sigma$ dashed lines (red), showing how outlier cases occur more frequently in financial markets (heavier tails) than a normal distribution. The large black swan events are typical of major economic events, in particular the Covid-crash circa 2020-21 and the U.S. tariff geopolitical situation in 2024.

2.2 Gain/Loss Asymmetry (Skewness)

Log returns were calculated as $r_t = \ln(1 + R_t)$, where R_t denotes the simple return. We measured the skewness, mean and median of each sample, using log returns.

Industry	Skewness	Mean	Median
Nondurable goods	-0.7359259	0.0003567811	0.0007996802
High-tech	-0.4080304	0.0006074447	0.001299156
Energy	-0.3217479	0.0002032179	0.000499875
Telecommunication	-0.5513389	0.0004153115	0.0006997551
Health	-0.4145099	0.0004153115	0.0009995003

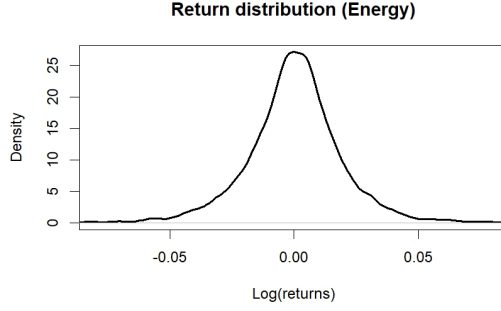
Table 3: Skewness across industries

As table 3 shows, all five industries had a negative skewness, which can be interpreted as: the sample contains more small gains and bigger losses. One reason for this can be psychological, as people tend to react more heavily to losses (sell their stocks in panic) than to equivalent gains.

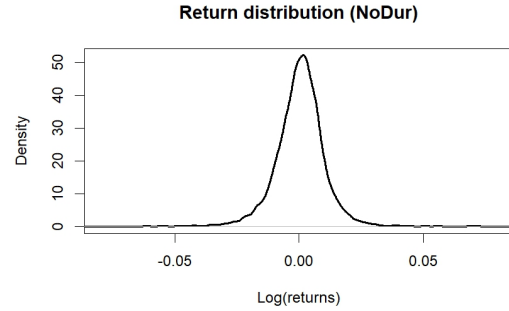
The least skewed is energy, where a reason for this can be that the energy markets are heavily dependent on natural resources (crude oil, natural gas), and so the market is influenced by political decisions. (Iran war/ closure of the Strait of Hormuz, or the invasion of Ukraine in 2022) In case of crude oil, there are the OPEC and the OPEC+, who in case of a price drop limit their production to push the price back or prevent further decline in the price.

In all cases the median is higher than the mean; it is in line with our expectation, that in every case the left tail is fatter or longer than the right one.

Due to the magnitude of the skewness, we cannot observe many things from the visualization, but according to the calculated data the first plot should be closer to symmetric with a skewness of approximately -0,32, while the second one should have fatter/longer left tails (compared to it's own right tail) with a skewness of approximately -0.74.



(a) Energy



(b) Nondurable goods

Figure 2: Log-Return distributions of Energy and Nondurable goods

2.3 Aggregational Gaussianity

The idea of Aggregational Gaussianity suggests that return distributions converge toward a normal distribution as the frequency of data aggregation decreases. Our results support this, as seen in table 4 there is a rapid decline in excess kurtosis across all sectors as we transition from daily data to longer periods.

	Enrgy	HiTec	NoDur	Telcm	Hlth
Daily	14.0671738	8.7888975	16.0701811	8.9049349	7.8482579
Weekly	6.1229800	2.1527153	9.4342695	4.6606163	2.6427235
Bi-Weekly	5.2516701	1.3795240	2.3416085	1.2240303	0.9523883
Monthly	4.8605162	-0.0702451	0.6900172	0.7079625	-0.2330436
Bi-Monthly	0.7639504	-0.7261987	0.1581366	0.1768149	-0.7324309
Yearly	-0.5094381	0.3948942	0.7072893	0.8065944	-1.3913588

Table 4: Excess kurtosis at different time frequencies

Information efficient and asset-light sectors like Technology and Consumer goods normalize quickly along with our baseline, appearing near Gaussian at even monthly intervals. Conversely the high-volatility, physically constrained sector "Energy" takes longer to reach normality. We can also see this visually in figure 3.

Economically, this isn't surprising: one cannot instantly drill more oil during a crisis. These physical supply chain bottlenecks take months to correct, keeping the distribution "non-normal" for a longer period compared to sectors like Healthcare, which we assume to be stable regardless of economic conditions. The persistence of these heavy tails

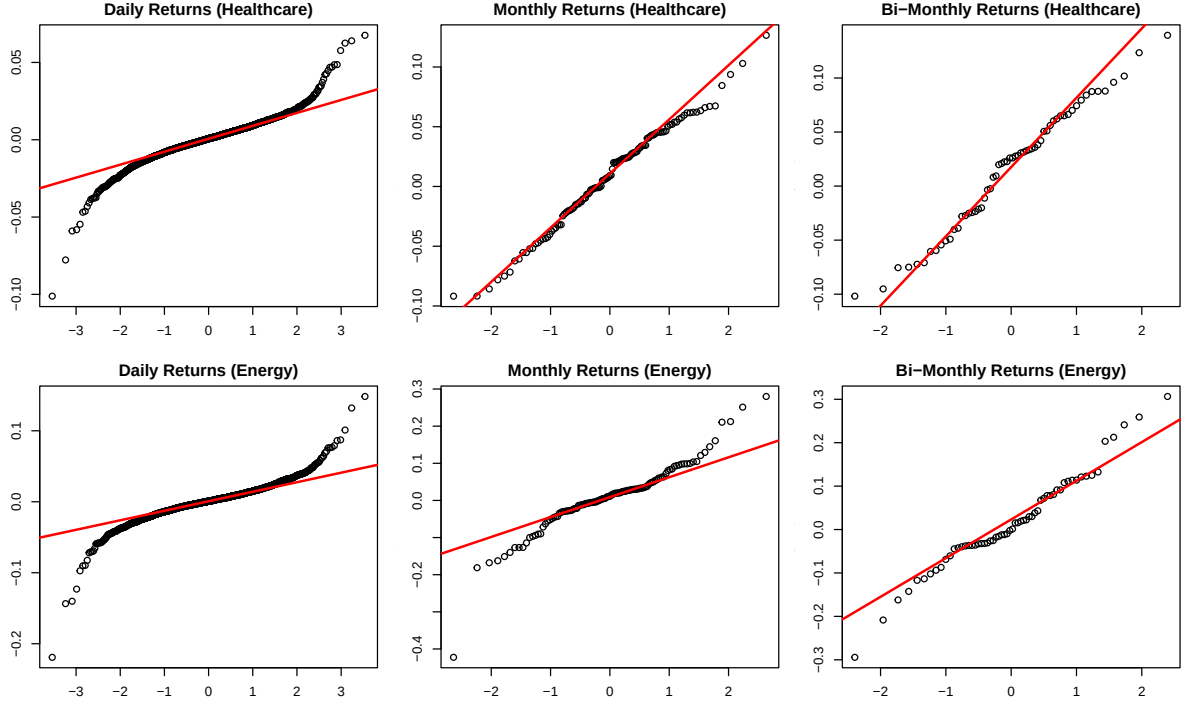


Figure 3: Aggregational Gaussianity in Healthcare and Energy

in Energy suggests that physical constraints are actively shaping market risk, rather than these stylized facts being purely predictable financial math. We are observing the difference between industries that react instantly to market signals and those held down by the hard reality of physical production.

However, there is a limitation to the theory. Our data shows that the real world doesn't perfectly follow the math. Instead of simply settling at normal levels, physically constrained or defensive sectors like Energy and Healthcare actually over-correct over a year. They drop from extreme daily highs (14.07 and 7.85, respectively) down to unnaturally stable, negative yearly levels (-0.51 and -1.39) as rigid demand and supply cycles force prices back to reality. Furthermore, this "smoothing out" effect isn't a straight line. Sectors like Technology and Non-Durables hit their most predictable, "normal" state around the one-to-two-month mark (dropping to near-zero levels of -0.07 and 0.15), only for extreme risks to spike again at the one-year horizon (rebounding to 0.39 and 0.71). This rebound exposes a critical limitation of the theory: while mathematical averaging can easily wash out daily trading noise, it cannot erase the real-world institutional calendar. Coordinated events like quarterly earnings, macroeconomic shifts, and year-end portfolio rebalancing inevitably reintroduce systemic shocks, proving that long-term market risks are ultimately driven by human behavior and physical realities rather than a continuous mathematical progression toward normality.

2.4 Volatility Clustering

This is the phenomenon where calculating volatility using a rolling window reveals a non-constant pattern. In other words, turbulent days cluster together (large price changes follow large price changes). Healthcare/Consumer shocks are digested by the market quickly. Physical supply chain shocks (like a destroyed pipeline or memory shortage)

take months to correct, so the “turbulence” should cluster longer in the Energy and Technology sectors. A standard way to measure this is by looking at the autocorrelation of squared returns. If volatility clusters, squared returns will exhibit significant positive autocorrelation that decays slowly.

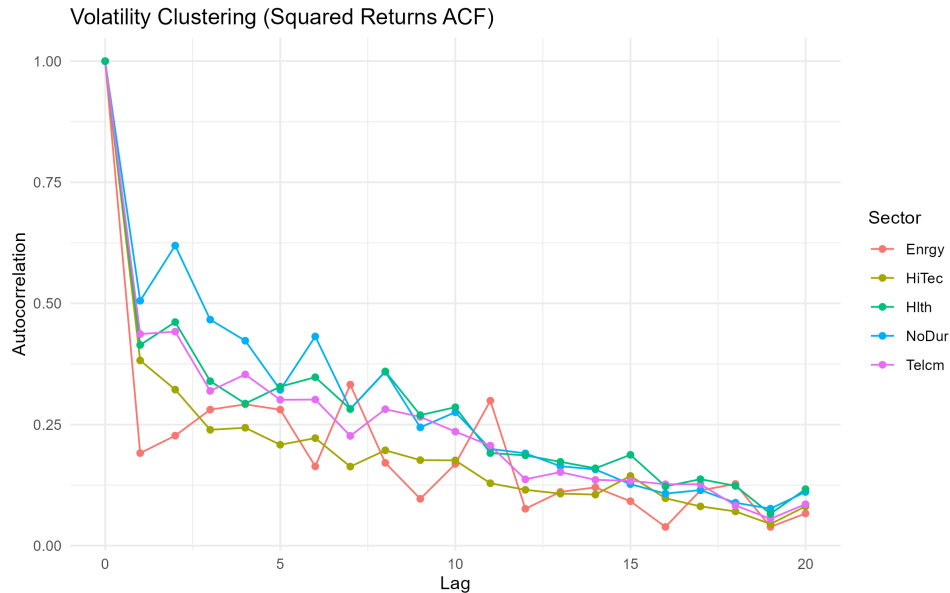


Figure 4: Autocorrelation of squared returns across sectors.

On the x-axis we have the lag: the time separation between data points, measured in days. On the y-axis we have autocorrelation: the relation between a series and the lagged version of that series. A very high autocorrelation at lag l means that in l days from now we can expect a price swing of a similar magnitude to the one we have today. at lag=0 all sectors have autocorrelation 1; in 0 days from now we can expect exactly the same price swing as today.

Firstly we note that Enrgy declines sharply at lag=1, and continues to experience relatively big shocks in autocorrelation, both upward and downward. We can interpret this as the sector being affected by many abrupt and unrelated shocks.

Telcm on the other hand declines much more smoothly in autocorrelation. This means that the sector is quite stable; there are no great disruptions that cause new spikes in autocorrelation, and we can say that there is some clustering.

NoDur has the highest autocorrelation of all sectors at lag=1, and continues to have the highest values for most lag values on the graph. Here we see the most clustering of all sectors.

Hitec has the smoothest and quickest decrease in autocorrelation. Shocks are quickly absorbed into the system.

Hlth does **not** have quickly declining autocorrelation. It takes longer for shocks to be ‘forgotten’ by the market.

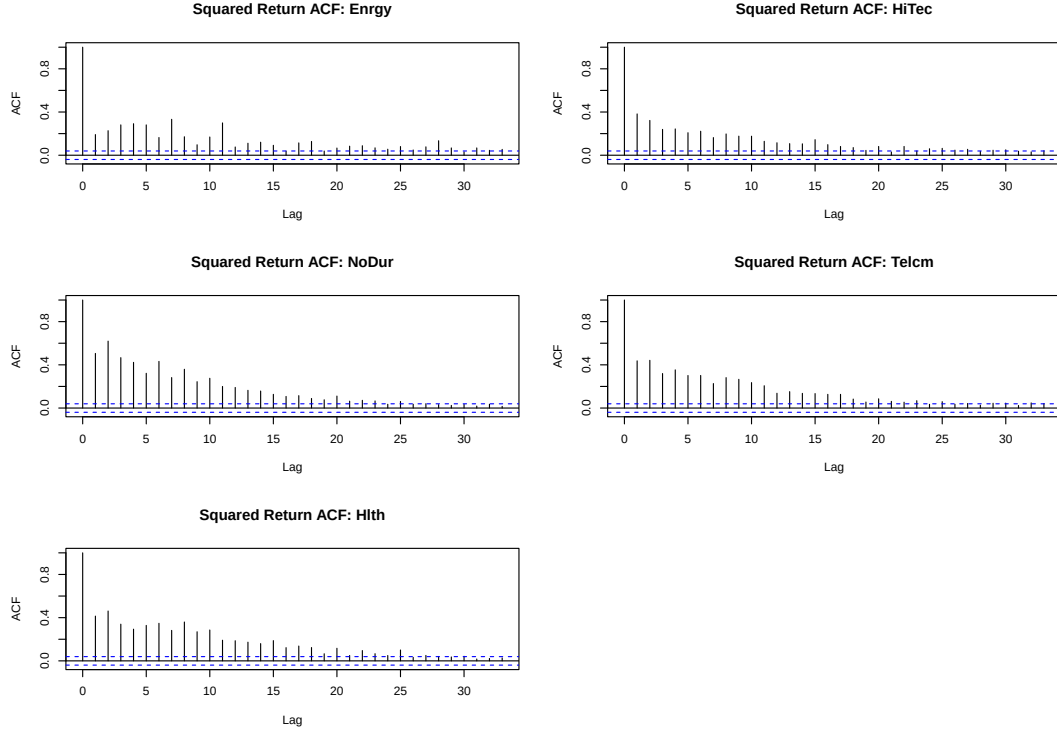


Figure 5: Autocorrelation functions of squared sector returns. Energy and Technology exhibit slower decay in autocorrelation, indicating stronger volatility clustering relative to Healthcare and Consumer Goods.

In this graph we still measure autocorrelation at lag x for each sector, but now we can also have a 95% confidence interval, represented by the blue lines.

From this we can see that Enrgy, even at higher lag still has significant spikes in autocorrelation. Again we can interpret this as the sector being prone to many unrelated shocks. NoDur has the most clustering. It has the highest values at small lag, and it takes the longest for the values to consistently fall within the confidence interval. This means that shocks to the market create great uncertainty, and take a long time to be forgotten.

2.5 The Leverage Effect vs. Inverse Leverage (Inventory) Effect

While volatility clustering shows the persistence of market turbulence, the leverage effect examines the direction that triggers the turbulence in the first place. Traditionally, equities exhibit a "Leverage Effect" where volatility spikes significantly when prices fall, as investors panic. We hypothesized that physically constrained markets like Energy might exhibit the opposite "Inventory Effect" where volatility spikes, on price surges due to fears of physical stock-outs.

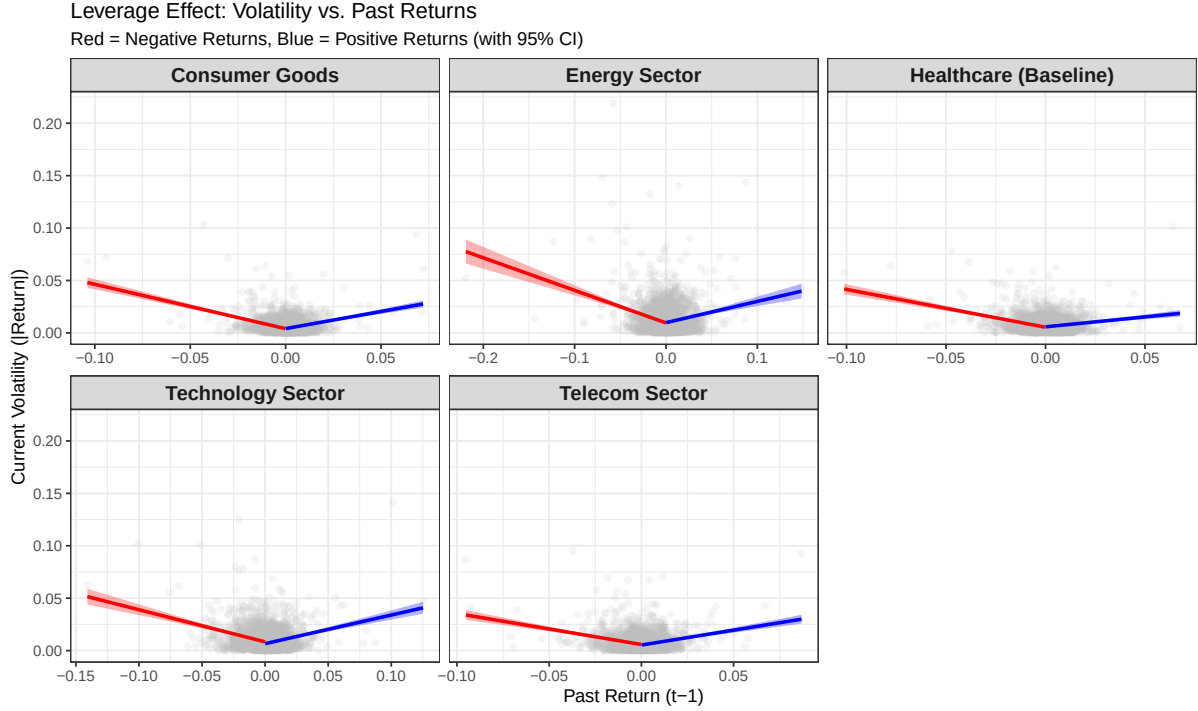


Figure 6: Leverage effect

Contrary to our hypothesis Figure 6 demonstrates a standard leverage effect across all observed sectors. The steeper slope of the negative returns compared to the positive returns is visibly present across the board, including in the Energy sector. To confirm this visual trend quantitatively, we calculated the exact Leverage Ratios defined as: $\frac{|Neg_Slope|}{Pos_Slope}$, where a ratio greater than 1.0 indicates a standard leverage effect (downside panic), while a ratio below 1.0 would indicate an inventory effect.

Sector	Neg_Slope	Pos_Slope	Leverage_Ratio
Consumer Goods	-0.4231946	0.3257909	1.298976
Energy Sector	-0.3105794	0.2025042	1.533694
Healthcare (Baseline)	-0.3605208	0.1897583	1.899894
Technology Sector	-0.3063558	0.2731609	1.121521
Telecom Sector	-0.2957982	0.2840360	1.041411

Table 5: Slope

As shown in Table 5, the Energy sector yielded a Leverage Ratio of 1.53, meaning negative returns amplify volatility 53% more violently than positive returns of the same magnitude. This rejects the presence of an Inventory Effect within this dataset.

We attribute this to the fundamental nature of the Kenneth French industry portfolios. Because these portfolios track the publicly traded equities of energy companies rather than the physical prices of the underlying commodities (e.g. Crude Oil), financial frictions overpower physical constraints. During broad market drawdowns, investors liquidate energy equities alongside the rest of the market, driving down prices and spiking volatility. The physical reality of the underlying supply chain is rendered secondary to the financialized reality of stock market mechanics.

Further supporting the dominance of financial over physical frictions is the behavior of

the Healthcare sector. Defined in our initial matrix (Table 1) as a stable, low-risk baseline, Healthcare exhibits the highest asymmetry of any sector analyzed, with a Leverage Ratio of 1.90. While its baseline volatility may be low, its reaction to negative shocks—such as failed clinical trials or sudden regulatory changes is extreme, highlighting how information processing and investor panic shape risk differently across sectors.

3 Conclusion

The results of our work suggest that industries exposed to stronger physical constraints (e.g. Energy) generally exhibit stronger persistence in volatility, and asymmetry patterns. This supports the argument of risk being shaped not only by financial dynamics, but also by economic conditions.

Volatility clustering appeared more persistent in sectors affected by supply-chain rigidities and slower adjustment mechanisms.

At the same time, the leverage effect’s persistence across all sectors points to investor reactions and market mechanics as a central driver of amplified risk. Overall, the results highlight that market volatility emerges from an interaction between financial frictions, human behavior, and real-world economic structures.

4 References

- Kenneth R. French Data Library
- Fan, Jianqing, and Qiwei Yao. *The Elements of Financial Econometrics*. Cambridge University Press, 2017.